

INDIA HAS AN ambitious target of generating 500 Gigawatt (GW) power from non-fossil fuels by 2030. That it is taking the target seriously is apparent from the 30% increase in allocation for renewable energy in the recently tabled Union Budget 2026-27 compared to FY26 revised estimates. But like most great stories, this one is also fraught with obstacles; this time, it's the volatility in metal prices.

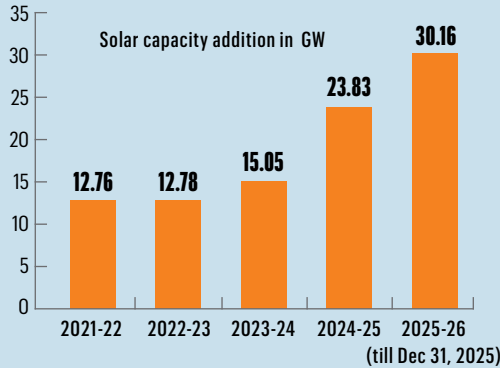
Some metals—notably silver, but also copper—are key inputs in photovoltaic (PV) cells, used to generate solar power. This is how it works: silver is converted into a paste and then loaded on a silicon wafer. When light strikes the silicon, it liberates electrons, and silver carries the generated electricity for immediate use or storage.

Alongside silver, metals such as copper and aluminium are used in wiring, busbars, connectors, and module frames. Utility scale solar manufacturing requires around three tonnes copper per MW capacity while onshore wind generation needs four tonnes per MW.

However, the sharp rise in silver and copper prices over the last few quarters are creating cost and supply headwinds for solar photovoltaic (PV) cell and module manufacturers. Silver has seen a record rally of 200% in the past 12 months. It has risen from around ₹60,000 to nearly ₹4,00,000 per kg. Copper has seen more than 45% jump in prices in the domestic market. The current raw material environment has reinforced the solar industry's focus on operational efficiency, disciplined sourcing, and technology-led optimisation, rather than reactive pricing adjustments.

CLEAN ENERGY DREAMS

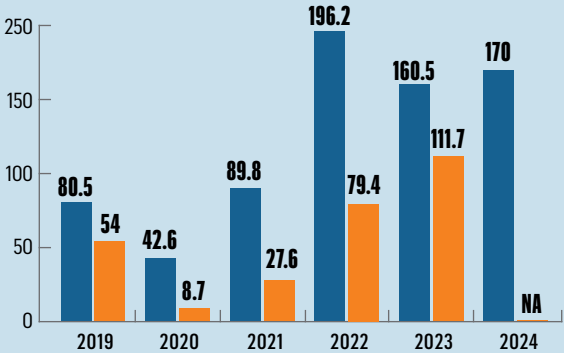
► Solar is set to be the largest contributor to India's 500GW non-fossil fuel energy target by 2030



SOURCE MINISTRY OF NEW AND RENEWABLE ENERGY

SILVER IMPORTS RISE

► Demand to remain high as solar scales up



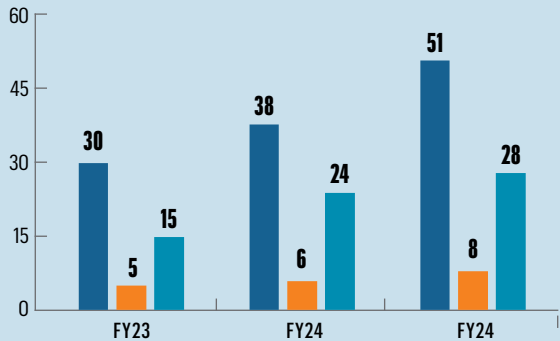
● TOTAL INDIA SILVER DEMAND (MOZ)
● SILVER IMPORTS (MOZ)

NOTE UNITS IN MILLION OUNCES (MOZ)
SOURCE WSS: WORLD SILVER SURVEY 2024

COPPER MAY KEEP SHINING

► Solar accounts for the highest demand for copper in the renewable power sector

► EVs are the second-largest demand drivers



● SOLAR ● RENEWABLE (EXCLUDING SOLAR)
● EV & CHARGING STATIONS

NOTE IN KT (KILO TONNES) **SOURCE** INTERNATIONAL COPPER ASSOCIATION INDIA